Multigranularity Selective Fine-Tuning of Vision Foundation Models for Coastal Remote Sensing Scene Classification

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***Abstract*—Large-scale coastal reclamation activities in recent decades have had a significant impact on the sustainability of marine and terrestrial ecosystems. Remote sensing scene classification (RSSC) plays a critical role in environmental monitoring and marine management; however, existing RSSC techniques are underdeveloped for coastal regions, especially in the study of intraclass and interclass properties among different types of coastal remote sensing images. A benchmark dataset dedicated to coastal scene interpretation is crucially missing. To address this gap, we construct a new coastal remote sensing scene dataset, termed CROSS-14, which contains 3925 images spanning 14 representative coastal use patterns. To effectively handle the complex heterogeneity of coastal scenes, we propose a multigranularity selective fine-tuning (MGSF) framework to adapt vision foundation models (VFMs) for classifying coastal remote sensing images. MGSF adopts a progressive classification network consisting of linear probing and selective fine-tuning. At the sample granularity level, an uncertainty-aware reweighting (UAR) strategy is designed to emphasize difficult and ambiguous samples based on linear probing results. At the feature level, dual-selective adapters (DSA) are introduced to adaptively select informative tokens and channels while suppressing redundant representations. Extensive experiments conducted on CROSS-14 and two public benchmarks under various settings demonstrate that MGSF achieves superior performance compared to state-of-the-art methods. Code and dataset are available at https://github.com/wangxinlu-y/MGSF**

***Index Terms*—Coastal remote sensing scene classification (RSSC), multigranularity selective fine-tuning (MGSF), vision foundation models (VFMs).**

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I. Introduction

In RECENT decades, land reclamation activities have proliferated globally, driven by the growing demand for industrial expansion, urbanization, and port infrastructure [1]. While generating considerable economic benefits, these large-scale and intensive human activities have also resulted in substantial ecological costs, such as the degradation of marine and terrestrial ecosystems [2]. In this context, high-precision and efficient remote sensing monitoring of coastal zones is essential for promoting sustainable marine resource utilization and protecting ecological environments. As a core method of interpretation, remote sensing scene classification (RSSC) has been applied to many fields, such as national governance [3], security surveillance [4], and resource management [5], [6].

Early approaches to RSSC typically relied on handcrafted features, such as color histograms [7] and texture and shape descriptors [8], which were combined with traditional classifiers such as support vector machines [9] and random forests [10]. With the rise of deep learning, modern neural networks such as convolutional neural networks (CNNs) [11], [12], [13] have brought remarkable progress to this field. Their hierarchical structure enables extraction of multiscale features, leading to significant improvements in classification accuracy. However, due to the inherent locality of convolutional operations, CNNs exhibit limitations in capturing long-range dependencies in images with complex spatial structures. To mitigate this issue, various strategies have been explored, such as multiscale feature fusion [14] and feature pyramid networks [15]. Despite these advances, performance in highly complex remote sensing scenes remains unsatisfactory.

In recent years, transformer-based models have gained increasing attention and have been widely applied to RSSC [16]. Unlike CNNs, vision transformers (ViTs) can model global contextual relationships more efficiently via self-attention mechanisms [17], which is crucial for holistic scene understanding. For example, Chen et al. [18] introduced the hierarchical feature fusion block into transformers and leveraged multiscale context to enhance performance. Ma et al. [19] designed a multiscale sparse cross-attention (MSCN) to integrate features from different levels while reducing the influence of irrelevant information. Yang et al. [20] further proposed an explainable multiscale transformer-based RSSC method that jointly exploits self-attention in

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spatial and frequency domains for enhanced global multiscale modeling.

More recently, vision foundation models (VFMs) pretrained on large-scale image datasets have been explored as powerful backbones for scene classification. For example, Jia et al. [21] proposed a visual prompt tuning scheme that keeps the ViT backbone frozen while optimizing task-specific visual prompts to achieve competitive performance. Chen et al. [22] developed AdaptFormer, which inserts lightweight adapters into the feed-forward networks of transformer blocks to enable efficient adaptation for visual recognition tasks. In addition, Zhu et al. [23] integrated prompt tuning into a metalearning framework to effectively adapt pretrained ViTs to RSSC. Dong et al. [24] further proposed a parameter-efficient fine-tuning framework for remote sensing foundation models, leveraging quantization adapters and context-aware prompts to further boost RSSC performance.

Although existing VFM-based methods have achieved strong performance on standard RSSC benchmark, directly applying them to coastal RSSC remains challenging due to the substantial domain gap between terrestrial and coastal scenes. Most publicly available RSSC datasets are designed for terrestrial land-use interpretation, and a dedicated benchmark for coastal scene classification is still lacking. In addition, compared to terrestrial scenes, coastal remote sensing images exhibit distinctive characteristics. For example, most of the area in the image is occupied by sea regions, while objects in the foreground area are sparsely distributed. This pronounced background redundancy and foreground sparsity significantly hinder feature discrimination and limit the effectiveness of existing models and fine-tuning strategies for coastal scene understanding.

To address these challenges, we first construct a new coastal RSSC dataset named CROSS-14, which consists of 3925 images covering 14 representative coastal use patterns. Then, we propose a multigranularity selective fine-tuning (MGSF) to effectively adapt VFMs for coastal RSSC tasks. MGSF selects informative samples, tokens, and channels at different granularities to progressively fine-tune VFMs. At the sample granularity level, an uncertainty-aware reweighting (UAR) scheme is developed to emphasize difficult samples that need more attention during model fine-tuning. At the feature level, a dual-selective adapter (DSA) operated on the reweighted samples is introduced into VFMs. The DSA jointly selects informative tokens and channels to preserve discriminative representations through an importance estimator (ImpEst) while suppressing redundant information using a sparsity controller (SpaCtrl). Finally, features from different granularities are integrated for the classification of coastal images. Extensive experiments on the CROSS-14 dataset and two public benchmarks, UC Merced (UCM) [25] and Northwestern Polytechnical University-RS Image Scene Classification 45 (NWPU-RESISC45) [26], demonstrate that MGSF consistently outperforms 14 state-of-the-art methods. Ablation experiments further verified the effectiveness of the proposed multigranularity selection mechanisms.

The main contributions of this article are summarized as follows.

1) We construct a new dataset, termed CROSS-14, which contains 3925 images and covers 14 coastal use patterns, providing a dedicated benchmark for fine-grained coastal scene classification.
2) We propose a MGSF framework that adapts VFMs to coastal RSSC tasks by selectively emphasizing informative samples, tokens, and channels at different granularities, thereby improving the efficiency and robustness of model fine-tuning.
3) We design a DSA that jointly evaluates and integrates representations at token and channel granularities to enhance discriminative features while reducing redundancy among them.
4) Experiments on CROSS-14, UCM, and NWPU datasets demonstrate that the proposed MGSF method achieves superior performance compared to some state-of-the-art methods.

The rest of this article is organized as follows. In Section II, we briefly review ViTs for RSSC and the fine-tuning strategies of VFMs. We introduce CROSS-14 in Section III and present the MGSF framework in Section IV. Subsequently, experiments and analysis of all methods are given in Section V. Finally, we conclude this article in Section VI.

II. Related Work

A. ViT for RSSC

Remote sensing scenes typically exhibit strong long-range dependencies because their semantic information is jointly defined by widely distributed and spatially separated objects [27]. Furthermore, objects in remote sensing images often present complex spatial layouts, with multiscale structures and arbitrary orientations. CNNs, despite their strengths, have inherent limitations in modeling such global context due to their local receptive fields. Transformers address these challenges more effectively through self-attention mechanisms that enable global dependency modeling.

Building on these advantages, Xu et al. [28] employed a ViT as a teacher to guide a lightweight CNN for RSSC via knowledge distillation, effectively integrating the global modeling ability of ViT with the efficiency of small networks. Lv et al. [29] introduced progressive label aggregation and a lightweight attention mechanism into ViT to better utilize structural and channel information in images. Following a similar paradigm, Wang et al. [30] incorporated pyramid pooling self-attention and a gating mechanism to model scene information along both spatial and channel dimensions. Chen et al. [31] combined a transformer and a CNN to jointly utilize local and global features. Subsequently, Zhao et al. [32] also designed a two-stream network consisting of a ViT and a CNN, in which calibration and joint supervision were considered for better classification. Tang et al. [33] developed an efficient multiscale transformer with cross-level attention learning (EMTCAL) to capture rich contextual information in images. Li et al. [34] developed a cotraining transformer to jointly achieve the classification, segmentation, and detection of remote sensing images. Ni et al. [35] designed a second-order differentiable token to adaptively select nonredundant

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tokens and obtain a more efficient representation for each image. Despite these advances, most existing Transformer-based RSSC methods treat samples and feature representations uniformly during training, without explicitly accounting for the heterogeneous importance of samples, spatial tokens, and feature channels—an issue that becomes particularly pronounced in complex and background-dominated coastal scenes [36].

B. Fine-Tuning of VFMs

In recent years, VFMs have exhibited outstanding representation ability and achieved remarkable improvement across a wide range of visual tasks [37]. Owing to their strong generalization ability, fine-tuning strategies have been developed to adapt VFMs to various downstream applications [38]. Among them, parameter-efficient fine-tuning methods have attracted increasing attention, as they significantly reduce training cost while preserving performance. They can be roughly divided into three different paradigms: prompt learning, reparameterization, and adapters. Prompt learning adapts VFMs by keeping the backbones frozen and optimizing a set of task-specific prompt parameters. Representative works include clustering-based prompt learning [39] and implicit prompt generation via shallow layers [40]. Reparameterization fine-tuning methods introduce extra trainable parameters to reparameterize the pretrained weights, such as adjusting deep features during training and reparameterizing them at inference time [41], or attaching low-rank residual branches to enable flexible parameter updates [42]. Adapter-based methods instead strategically insert lightweight bottleneck networks into selected blocks of VFMs and train only these modules for downstream tasks. For example, Chen et al. [43] proposed a pretraining-free dense prediction adapter incorporating spatial prior and injection, while Jie and Deng [44] introduced lightweight convolutional bypass modules into multilayer perceptron layers to inject visual inductive biases.

Besides, VFMs and their fine-tuning strategies have also been explored in the remote sensing field [45]. He et al. [46] proposed a VFM adaptation framework that keeps the backbone frozen while introducing cross-spatial and cross-channel interaction modules for multimodal remote sensing classification. Wang et al. [47] advanced a remote sensing foundation model by pretraining a ViT using masked autoencoders. Wu et al. [48] proposed a semantic-enhanced multimodal remote sensing foundation model that uses progressive pretraining to achieve strong generalization across diverse Earth observation tasks. These approaches demonstrate that parameter-efficient fine-tuning methods can substantially reduce model size and computational complexity while maintaining competitive performance [49]. This limitation becomes particularly critical for coastal remote sensing scenes, where significant intraclass variance and interclass similarity coexist, with large homogeneous sea regions dominating the background, and informative foreground objects being sparse and spatially dispersed. Such spatial imbalance and pronounced heterogeneity significantly hinder effective feature discrimination, rendering vanilla fine-tuning strategies suboptimal for coastal RSSC. To address these challenges, we propose the MGSF framework that explicitly models contribution at the sample, token, and channel granularities. By integrating UAR and DSA, MGSF selectively emphasizes informative samples and representations while suppressing redundant background information, enabling more effective adaptation of VFMs for coastal RSSC.

III. CROSS-14 DATASET

A. Dataset Construction

Coastal RSSC plays an important role in marine ecological protection and coastal zone management and has become an indispensable component of Earth observation applications. Nevertheless, most publicly available RSSC benchmarks are primarily designed for terrestrial land-use interpretation, where the dominant categories are urban, agricultural, or forest scenes. Although a limited number of coastal remote sensing images are included in these datasets, there is still no dedicated benchmark that systematically analyzes diverse coastal use patterns in a fine-grained manner.

To fill this gap, we develop the CROSS-14 dataset aiming to establish a new benchmark for coastal RSSC.

Remote sensing images covering the entire Chinese coastal zone are collected from Google Earth Engine, with a spatial resolution of 1 m. After data curation and quality control, a dataset of 3925 coastal remote sensing images with a size of 640×640 pixels is selected. Guided by established descriptions and literature on coastal utilization typologies [50], [51], the images were annotated to identify 14 distinct coastal use patterns. The annotation process was conducted by two researchers with experience in remote sensing image interpretation. Each image was independently reviewed by both annotators according to predefined category descriptions. Samples with inconsistent labels were re-examined and assigned final labels through discussion between the two annotators. Images that remained visually ambiguous or lacked sufficient discriminative information were excluded during quality control. These patterns include: 1) artificial island extraction; 2) bathing beach; 3) coastal aquaculture; 4) construction reclamation; 5) cross-sea bridge; 6) harbor; 7) impermeable structure; 8) navigation; 9) open aquaculture; 10) permeable structure; 11) platform extraction; 12) recreation; 13) saltpans; and 14) water intake and outfall. Each category contains between 63 and 603 images. Fig. 1(a) shows the spatial coverage in this dataset and provides representative examples for each class.

B. Dataset Statistics

Illustrated in Fig. 1(b), the CROSS-14 dataset contains five primary coastal use types, which are further refined into 14 fine-grained classes. Different from terrestrial benchmarks that group scenes by dominant land-cover textures, we categorize CROSS-14 based primarily on coastal use patterns and their spatial relationship to sea and shorelines. Fig. 1(c) further reports the number of samples per category, from which class imbalance can be observed.

Beyond these basic statistics, the CROSS-14 dataset also exhibits several distinctive characteristics that make it particularly challenging. First, due to the fine-grained annotation,

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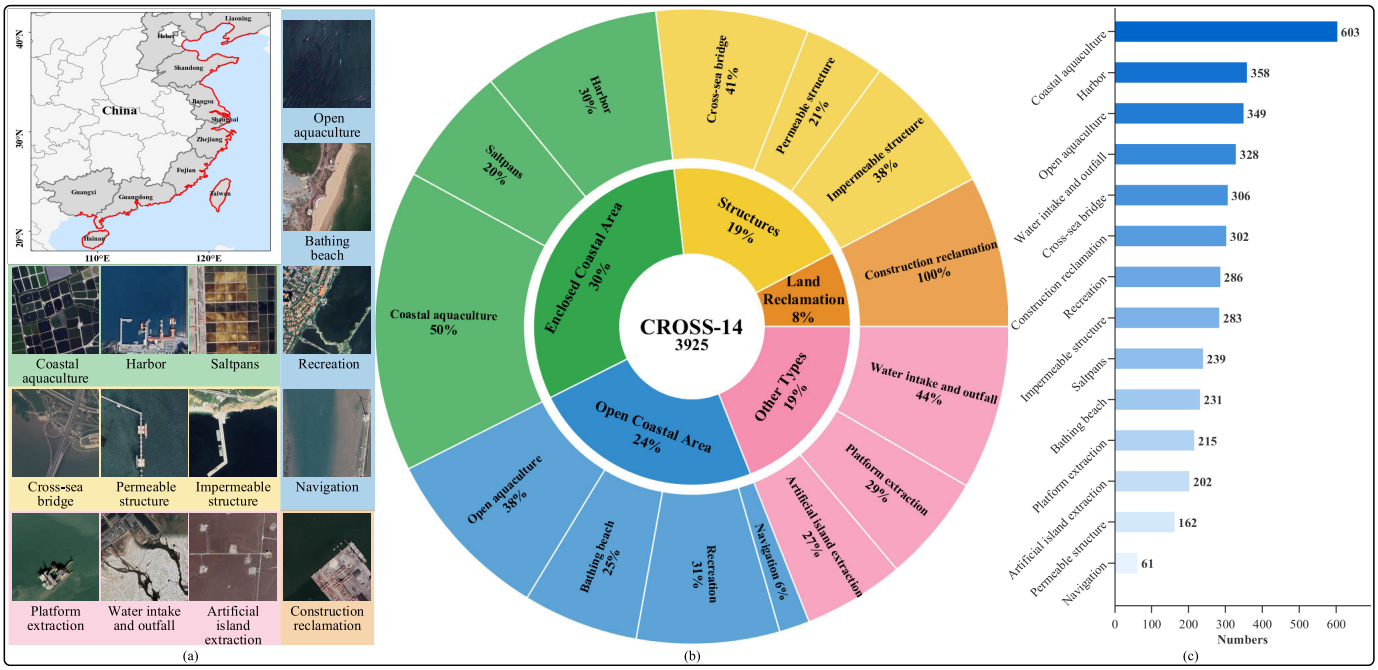


Fig. 1. Overview of the proposed CROSS-14 dataset. (a) Data coverage and representative category examples, (b) proportions of different coastal use patterns, and (c) data distribution of different patterns in the dataset.

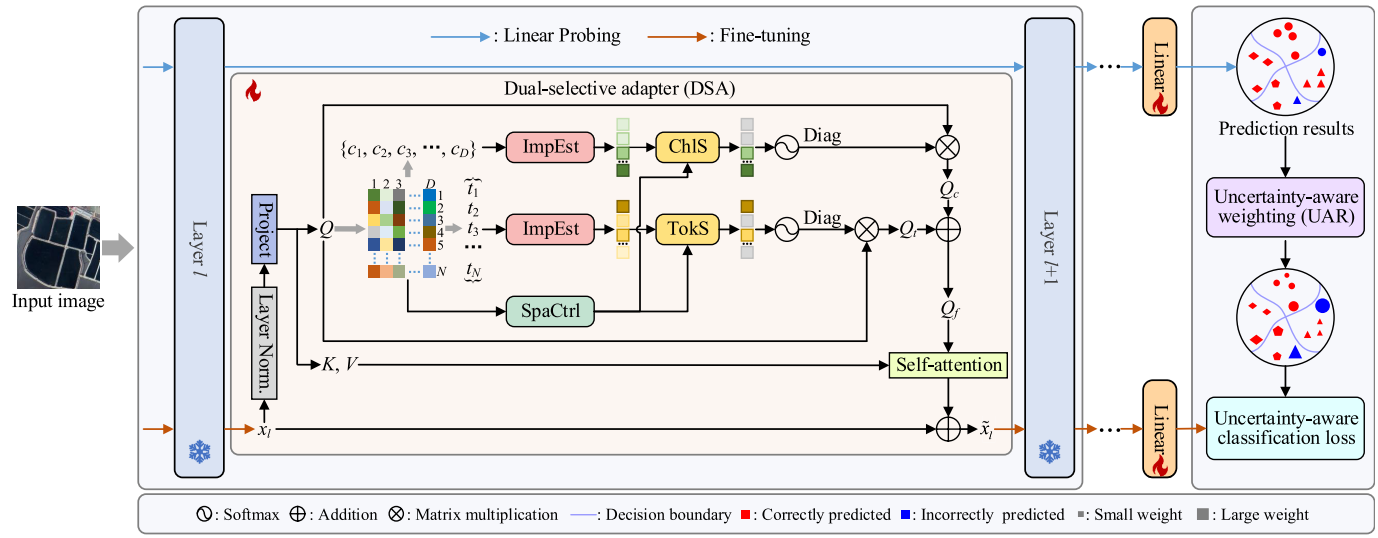


Fig. 2. Overall architecture of the proposed MGSF.

there is significant similarity across scenes belonging to different categories. For example, coastal aquaculture and open aquaculture have similar visual appearances. Similar ambiguity also exists in the data for impermeable and permeable structures. Second, due to variations in geographic location, hydrodynamic conditions, and construction stages, the dataset exhibits substantial intraclass heterogeneity in color, texture, and surrounding context even within the same coastal use pattern. Third, most images contain a large proportion of homogeneous sea surfaces, while the informative foreground facilities occupy only a small fraction of the scene but exhibit complex shapes, multiscale structures, and arbitrary orientations. Although scene semantics in both terrestrial and coastal remote sensing images are typically determined by a subset of informative regions, this property is significantly amplified in coastal remote sensing images because they exhibit stronger background redundancy and sparser, more scattered discriminative foreground. All these factors make the CROSS-14 dataset a challenging yet realistic benchmark for evaluating whether VFMs and their fine-tuning strategies can effectively handle the interclass and intraclass heterogeneity inherent in the coastal RSSC task.

IV. Proposed Method

Fig. 2 illustrates the overall workflow of the proposed MGSF framework. At the sample level, a linear probing

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strategy is first employed, where the backbone of the VFM is frozen and only the classification head is trained to analyze the characteristics and difficulty of training samples. Then, based on the uncertainty of prediction results of training samples, we design the UAR mechanism to estimate sample importance and highlight difficult samples during fine-tuning. Next, we incorporate the DSA into the VFM to adaptively select representative features at both the token and channel granularities. In DSA, ImpEst measures the importance of tokens and channels, while SpaCtrl is used to preserve informative features while effectively suppressing redundancy. Finally, the VFMs are effectively fine-tuned for coastal RSSC by combining uncertainty-induced classification loss and the selectively refined features. Sections IV-A–IV-C describe each component of the proposed framework in detail.

A. Linear Probing

During the linear probing stage in Fig. 2, the backbone of the VFMs is frozen, and only the linear classification head is trained. This strategy enables a lightweight yet effective adaptation to the RSSC of coastal images. In particular, each image is first fed into the frozen backbone to extract the latent feature representation, which is then mapped to class logits via a learnable linear classification head consisting of a single fully connected layer. By minimizing the cross-entropy loss on the training set, the classification head is optimized while preserving the general representation learned during pretraining. After the preliminary training is completed, we apply the frozen backbone together with the learned classification head to the entire training set and obtain the predicted label and the corresponding confidence distribution of each sample. The classification head learned during the linear probing stage is retained and used to initialize the classification head in the subsequent selective fine-tuning stage. During this stage, the reused classification head is jointly optimized with the inserted DSA modules, while the pretrained backbone remains frozen.

Through linear probing, these sample-level predictions provide valuable insights into the intrinsic properties of the training data, such as classification difficulty, semantic ambiguity, and feature discriminability. This is particularly important for the RSSC task, where large intraclass variation and high interclass similarity often coexist, making uniform fine-tuning suboptimal. While correct and incorrect predictions demonstrate the difficulty of classification, the associated confidence scores of these predictions further reveal the degree of semantic certainty or ambiguity underlying each sample. These sample-granularity cues provide an initial characterization of the training set, which serves as a reliable basis for subsequent customized weight estimation in the UAR strategy.

B. Uncertainty-Aware Reweighting

As discussed above, the preliminary sample-granularity prediction results in the linear probing stage contain rich information about the intraclass and interclass properties of the training data. The predicted labels indicate classification correctness, while the associated confidence scores quantify prediction uncertainty. Based on these two factors, these

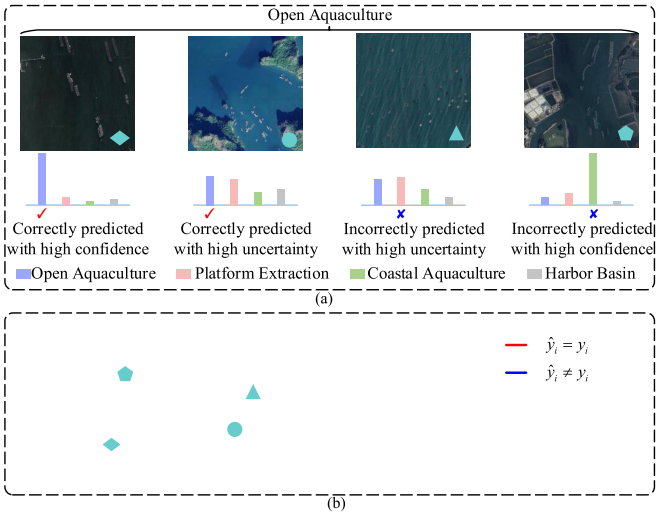


Fig. 3. UAR mechanism. (a) Predicted labels and confidence of different samples. (b) Uncertainty-aware weights of different samples.

samples can be coarsely divided into four types, as shown in Fig. 3(a), to depict their classification difficulty. High uncertainty indicates that the interclass discriminability is low due to semantic ambiguity among different scenes. Such heterogeneous prediction behaviors suggest that different samples contribute unequally to model fine-tuning. Prediction correctness and uncertainty characterize sample difficulty from complementary perspectives. Using uncertainty alone cannot distinguish between correctly classified boundary samples and misclassified ambiguous samples, although they may exhibit similar entropy values. More importantly, entropy-only weighting may overlook high-confidence misclassified samples, which have low entropy but indicate systematic prediction errors that require stronger correction. On the contrary, misclassification-based weighting provides only a binary distinction and treats all incorrect samples equally, regardless of whether the error is confident or highly ambiguous. This may overemphasize uncertain samples that provide less reliable optimization signals. Consequently, jointly considering prediction correctness and uncertainty enables a finer characterization of training samples and provides a more reliable basis for differentiated weighting across the four prediction cases. Therefore, we should assign distinct weights to samples according to their attributes, enabling the fine-tuned models to focus more effectively on difficult and ambiguous cases.

Motivated by the above observations, we propose the UAR strategy that leverages both the predicted labels and their associated confidence scores. In particular, we calculate the entropy of the confidence scores of each sample to measure its uncertainty

$$u_i = -\sum_{c=1}^{C} p_i^c \log p_i^c \tag{1}$$

where the subscript i denotes the sample index. C stands for the number of classes and p_i^c is the confidence of the c th class. In general, high uncertainty corresponds to moderate confidence scores, indicating increased semantic ambiguity.

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Then, considering the predicted label, the UAR function is defined as follows:

$$w_i = \begin{cases} 1 - e^{-u_i/t_1+\alpha}, & \hat{y}_i = y_i \\ 1 + e^{-u_i/t_2+\beta}, & \hat{y}_i \neq y_i \end{cases} \tag{2}$$

where w_i is the weight of the ith sample. $\hat{y}_i$ and y_i correspond to the predicted label and ground truth, respectively. The parameter t_1 and t_2 are temperature coefficients that control the sensitivity of the weighting function to uncertainty, while α and β are hyperparameters that control the baseline and growth rate of the weights.

Fig. 3(b) illustrates the UAR function and the computed weights for representative samples. For correctly classified samples with high confidence, weights are reduced to avoid redundant learning. Correctly classified samples with high uncertainty are moderately upweighted to refine decision boundaries. Meanwhile, misclassified samples with high uncertainty are upweighted more gently, allowing the model to learn from ambiguous cases without introducing excessive noise. For misclassified samples with high confidence, their weights are substantially increased, imposing a strong penalty to correct confident errors.

By incorporating the sample-level weights into the standard cross-entropy loss, the uncertainty-aware cross-entropy (UACE) loss is defined as follows:

$$\mathcal{L}_{UACE} = \frac{1}{N}\sum_{i=1}^{N} w_i CE\left(\hat{y}_i, y_i\right) \tag{3}$$

where CE is the cross-entropy between the predicted result and the ground truth. N denotes the total number of samples. Notably, UACE reduces to the standard cross-entropy loss when uniform weights are applied. The proposed UACE loss is used for fine-tuning VFMs, enabling both the inserted DSA and the reused linear classification head to focus more on samples with high uncertainty. By modulating the contribution of different samples during optimization, the proposed loss promotes more discriminative feature learning for heterogeneous coastal remote sensing scenes.

C. Dual-Selective Adapter

Beyond the sample-granularity importance modeled by UAR, the learned features also exhibit substantial differences at both the token and channel granularities. In coastal remote sensing scenes, most tokens correspond to homogeneous sea backgrounds, whereas only a small subset captures discriminative foreground structures, resulting in significant redundancy during fine-tuning. At the channel granularity, different channels encode distinct semantic attributes, yet only a subset of them is informative for distinguishing fine-grained coastal land-use patterns. However, most existing sparse attention or token selection methods focus on token importance and rely on manually predefined top-K ratios, which limits their adaptability across different scenes [19]. Furthermore, importance differences among channels are often neglected, leading to redundant or noisy feature dimensions being retained.

To address these limitations, we design our DSA architecture to support a joint and adaptive modeling of importance at both token and channel granularities. As shown in Fig. 2, the proposed DSA integrates complementary selection mechanisms at the channel and token granularities and is positioned between two consecutive transformer blocks of a pretrained VFM. The detailed design of each component is described below.

TABLE I
IMPEST AND SPACTRL DETAILS

Network	Layer	Input	Output
ImpEst	Linear	D or N	32
	GELU	–	–
	Linear	32	32
	GELU	–	–
	Linear	32	1
SpaCtrl	Linear	$D \times N$	512
	LeakyReLU	–	–
	Linear	512	2
	sigmoid	2	2

1) Channel-Granularity Selection Mechanism: Channel-granularity selection aims to identify and emphasize informative feature dimensions while suppressing redundant or noisy ones. In DSA, the input x_l is first normalized and projected to obtain the query Q, key K, and value V. Given the query $Q \in \mathbb{R}^{N\times D}$, where N and D denote the numbers of tokens and channels, respectively, the ith channel across all tokens is aggregated to form a channel descriptor c_i. The resulting channel vectors c_1 through c_D are then fed into ImpEst to obtain the corresponding channel-wise score $s_c \in \mathbb{R}^D$

$$s_c = \text{ImpEst}\left(c_1, c_2, c_3, \ldots, c_D\right) \tag{4}$$

where ImpEst is implemented as a multilayer perceptron (MLP) with the architecture specified in Table I.

In order to let the model automatically decide how many channels should be preserved at each layer, we further introduce the SpaCtrl as shown in Table I, which predicts a channel retention ratio r_c to control the sparsity level of channels. Then, informative channels are selected according to r_c

$$\begin{aligned} w_c &= Softmax\left(s_c \cdot \mathbb{I}\left(s_c \in \text{Top-}K\right)\right) \\ K &= round\left(D \cdot r_c\right) \end{aligned} \tag{5}$$

where $\mathbb{I}$ is the channel selector (ChlS), also an indicator function. Finally, we reshape w_c as a diagonal matrix $W_c \in \mathbb{R}^{D\times D}$, which is applied to the query features to perform channel-wise reweighting

$$Q_c = QW_c. \tag{6}$$

Overall, the channel-granularity selection calibrates the feature subspace by assigning importance to embedding dimensions under a layer-adaptive sparsity budget. This preserves more discriminative channels and enhances semantic cues while filtering out noisy or irrelevant activations to produce a more compact representation.

2) Token-Granularity Selection Mechanism: Unlike channel selection, the token-granularity selection mechanism is designed to identify representative spatial regions and suppressing redundant background tokens. Given the same query

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$Q \in \mathbb{R}^{N\times D}$, ImpEst is applied to all tokens $\{t_1, t_2, t_3, \ldots, t_N\}$ to estimate their corresponding importance. Then, the token retention ratio r_t from SpaCtrl is utilized to filter out the trivial tokens. The corresponding token selector is denoted as TokS in Fig. 2. The retained important tokens are reweighted and rearranged as a diagonal matrix $W_t \in \mathbb{R}^{N\times N}$. Token selection is achieved by premultiplying the query with the reweighting matrix

$$Q_t = W_t Q. \tag{7}$$

Similar to channel-granularity selection, the token-granularity selection refines discriminative spatial components by dynamically estimating token saliency and selecting a variable subset of informative regions. This mechanism concentrates modeling capacity on salient regions while reducing the influence of redundant background tokens, leading to more relevant spatial cues for fine-tuning.

Finally, for joint learning between token and channel granularities, Q_c and Q_t are integrated to compute a fused query Q_f by

$$Q_f = \lambda Q_c + (1-\lambda)\, Q_t \tag{8}$$

where $\lambda \in (0, 1)$ is a learnable parameter. Then, Q_f is further combined with the key and value via self-attention to produce the output of DSA $\tilde{x}_l$

$$\tilde{x}_l = x_l + Softmax\left(\frac{Q_f K^T}{\sqrt{D}}\right) V. \tag{9}$$

Through the proposed DSA, the universal representations of VFMs are preserved while adaptive selection is injected into the hierarchical backbones. In this way, DSA progressively reweights important tokens and informative channels, enabling more fine-grained and effective fine-tuning for coastal RSSC.

V. Experiments

A. Datasets

This study evaluates the effectiveness of the proposed MGSF framework using the newly constructed CROSS-14 dataset and two widely used public remote sensing benchmarks. Section III provides a detailed description of CROSS-14, and below, we provide an overview of the two public datasets.

1) *UCM Dataset:* The UCM dataset [25] was constructed and released by the UCM computer vision laboratory. The dataset contains 21 classes with 100 images per class, resulting in a total of 2100 images. Each image has a spatial resolution of 0.3 m and a fixed size of 256 × 256 pixels. The dataset covers a diverse range of land-use categories, including: 1) agriculture; 2) airplanes; 3) baseball diamonds; 4) beaches; 5) buildings; 6) chaparral; 7) dense residential areas; 8) forests; 9) freeways; 10) golf courses; 11) harbors; 12) intersections; 13) medium-density residential areas; 14) mobile home parks; 15) overpasses; 16) parking lots; 17) rivers; 18) runways; 19) sparse residential areas; 20) storage tanks; and 21) tennis courts.
2) *NWPU Dataset:* Curated by Northwestern Polytechnical University, the NWPU dataset [26] contains 31 500 remote sensing images from 45 scene classes, with 700 images per class. All images are 256 × 256 pixels, and the spatial resolution ranges from 0.2 to 30 m. The 45 categories include: 1) airplane; 2) airport; 3) baseball diamond; 4) basketball court; 5) beach; 6) bridge; 7) chaparral; 8) church; 9) circular farmland; 10) cloud; 11) commercial area; 12) dense residential; 13) desert; 14) forest; 15) freeway; 16) golf course; 17) ground track field; 18) harbor; 19) industrial area; 20) intersection; 21) island; 22) lake; 23) meadow; 24) medium residential; 25) mobile home park; 26) mountain; 27) overpass; 28) palace; 29) parking lot; 30) railway; 31) railway station; 32) rectangular farmland; 33) river; 34) roundabout; 35) runway; 36) sea ice; 37) ship; 38) snowberg; 39) sparse residential; 40) stadium; 41) storage tank; 42) tennis court; 43) terrace; 44) thermal power station; and 45) wetland.

B. Implementation Details

In the following experiments, we fine-tune three VFMs for RSSC, including DeiT [52], T2T-ViT [53], and ViT [17]. For all three models, the base variants are employed, and their backbones are initialized with pretrained weights from ImageNet-21k. All experiments are conducted on a workstation with an NVIDIA GeForce RTX 3060 GPU. During the linear probing stage, the backbone network is frozen, and only the linear classification head is trained for 100 epochs, using a batch size of 512 and a learning rate of 0.001. In the second stage, the DSA modules are trained for 150 epochs with a batch size of 32, while the learning rate and weight decay are both set to 0.0001. In both stages, the AdamW optimizer is adopted for model optimization. For UAR estimation, the hyperparameters are set to $\alpha = -0.2$, $\beta = -0.05$, $t_1 = 2.0$, and $t_2 = 1.5$. These values are determined based on the performance analysis of ViT in Section V-G and are directly adopted for the remaining VFM backbones in subsequent experiments. All images of the dataset are randomly assigned to either the training set or the test set. The proportions of training data in CROSS-14 and UCM are 50% and 80%. The proportions of training data in NWPU are 10% and 20%, respectively. Each experiment is repeated five times and the mean and standard deviation of the results are reported. The performance of the method is quantitatively evaluated using overall accuracy (OA) and confusion matrix (CM). The OA indicates the number of correctly classified images divided by the total number of images. The CM is used to represent the probability that each category is predicted to belong to the other categories.

C. Comparison Experiments

To further validate the effectiveness of MGSF, we select a variety of representative baseline methods for comparison under consistent evaluation protocols. In particular, two CNN-based models, VGG-16 [54] and ResNet-50 [55], are included as classical baselines. Among VFMs, we employ ViT [17] and its variants or fine-tuning networks, including DeiT [52], Swin-T [56], T2T-ViT [53], AdaptFormer [22], PoolFormer

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TABLE II
COMPARISON RESULTS (%) OF ALL METHODS ON THREE DATASETS

Category	Model	CROSS-14		UCM		NWPU	
		50%	80%	50%	80%	10%	20%
CNN-based	VGG-16	83.68±0.12	85.58±0.14	96.18±0.60	98.11±0.89	88.17±0.18	91.62±0.19
	ResNet50	85.60±0.28	86.85±0.23	96.88±0.66	98.23±0.49	90.76±0.47	93.11±0.15
VFMs	Swin-T	86.68±0.13	88.32±0.12	99.14±0.10	99.46±0.15	90.84±0.14	93.18±0.19
	AdaptFormer	80.16±0.22	81.44±0.15	99.02±0.33	99.50±0.28	90.99±0.15	93.21±0.20
	PoolFormer	79.66±0.11	86.00±0.25	98.29±0.46	99.02±0.17	91.38±0.11	93.59±0.18
	FPET	86.16±0.17	88.25±0.16	99.04±0.12	99.76±0.14	92.26±0.11	95.06±0.14
RSSC-specific	CDLNet	86.37±0.07	88.06±0.04	99.06±0.15	99.71±0.09	93.43±0.18	95.30±0.03
	EMSCNet	82.59±0.11	86.13±0.14	99.10±0.20	99.71±0.17	93.58±0.22	95.37±0.07
	MSCN	84.10±0.07	88.28±0.09	99.26±0.07	99.80±0.10	92.64±0.09	94.59±0.11
	EMTCAL	86.62±0.21	88.88±0.14	98.67±0.16	99.57±0.28	91.63±0.19	93.65±0.12
	CGFNet	86.97±0.19	88.51±0.15	98.48±0.21	99.57±0.23	92.24±0.22	94.42±0.10
Full-parameter training	DeiT	83.30±0.24	86.07±0.21	98.60±0.25	99.29±0.33	93.10±0.24	95.30±0.11
	T2T-ViT	86.12±0.15	88.41±0.07	98.39±0.63	99.10±0.10	91.32±0.18	93.89±0.10
	ViT	84.07±0.18	87.93±0.11	98.75±0.21	99.29±0.34	91.59±0.08	93.90±0.17
Fine-tuning with MGSF	DeiT+MGSF	86.98±0.23	88.97±0.20	99.35±0.15	99.76±0.14	93.52±0.04	95.58±0.08
	T2T-ViT+MGSF	87.02±0.11	88.99±0.12	99.42±0.04	99.76±0.08	94.24±0.07	95.42±0.12
	ViT+MGSF	**87.13±0.14**	**89.02±0.11**	**99.43±0.13**	**99.90±0.10**	**94.54±0.10**	**95.94±0.11**

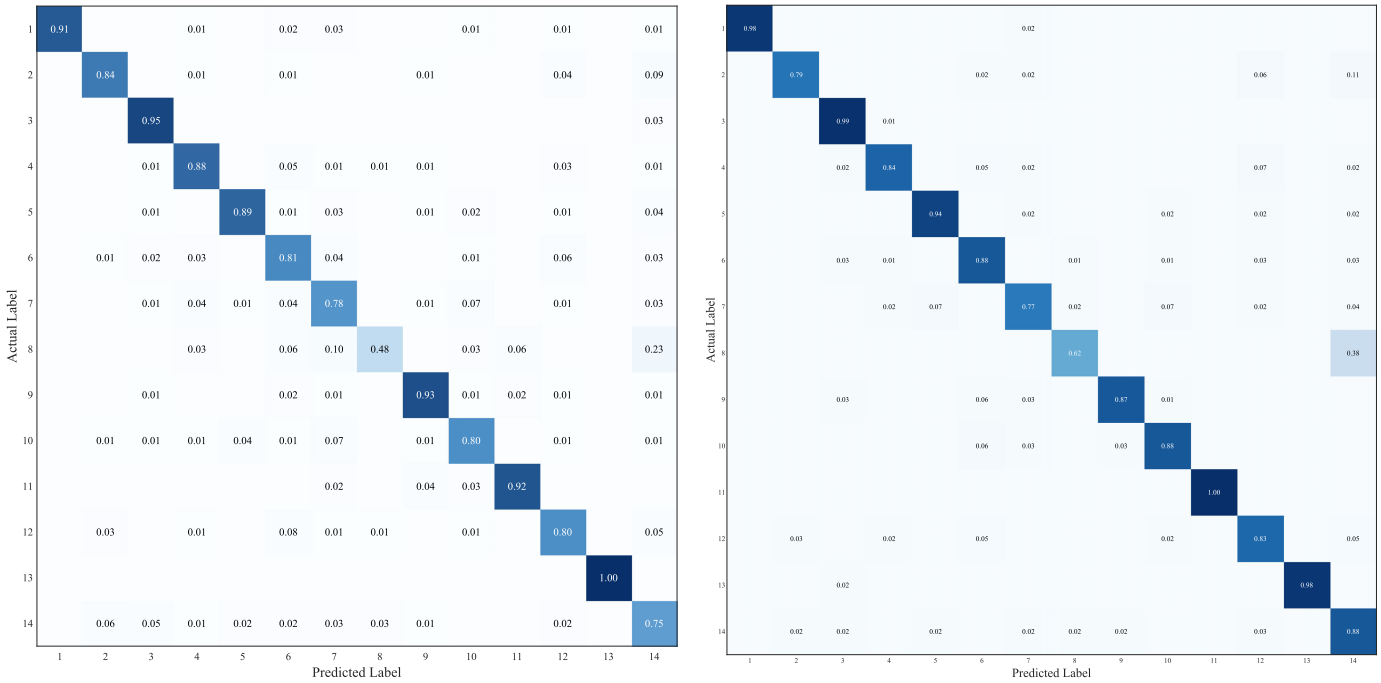
[57], and FPET [58], for comparative evaluation. In addition, we incorporate five state-of-the-art RSSC-specific methods for a more comprehensive comparison, including EMTCAL [33], efficient multisample contrastive network (EMSCNet) [59], collaborative dictionary learning network (CDLNet) [60], CGFNet [61], and MSCN [19]. All the aforementioned models are trained and evaluated on UCM, NWPU, and CROSS-14. All competing methods are retrained following the experimental configurations reported in their original papers or official implementations, including the input image size, data preprocessing, optimization settings, and training schedule. The same dataset partition ratios and evaluation metrics are used to ensure consistent evaluation across methods. Table II presents the classification results of all methods across different datasets. The best results are highlighted in bold, the second-best results are underlined, and the third-best results are marked with a dashed underline.

1) Comparison on Cross-14: For experiments on the CROSS-14 dataset, two training ratios—50% and 80%—are considered, with the remaining samples reserved for testing. From Table II, we can see that the DeiT, T2T-ViT, and ViT models fine-tuned with the proposed MGSF framework consistently achieve better performance than other methods. Among CNN-based methods, ViT+MGSF achieves OA improvements of 3.45% and 3.44% over VGG-16 on different data partitioning settings and achieves OA gains of 1.53% and 2.17% over ResNet-50, respectively. These results highlight the advantage of transformer-based architectures in modeling global contextual information. For example, Swin-T achieves an OA of 86.68% with 50% training samples, and T2T-ViT reaches an OA of 88.41% with 80% training samples. Despite the good performance of these methods, the combination of ViT and MGSF consistently delivers the best results, outperforming Swin-T and T2T-ViT by 0.45% and 0.61% in OA, respectively. For RSSC methods, EMTCAL achieves 88.88% accuracy when 80% training samples are used, whereas ViT+MGSF yields a 0.14% improvement. These gains can be attributed to the multigranularity selective strategy, which jointly models importance at the sample-, channel-, and token-granularity, thereby effectively suppressing homogeneous background regions and highlighting discriminative objects. Furthermore, it is worth noting that the performance of DeiT+MGSF and T2T-ViT+MGSF also outperforms their corresponding baseline methods, which also verifies the robustness of MGSF against the selection of hyperparameters.

The CMs of ViT+MGSF on the CROSS-14 dataset are illustrated in Fig. 4. Overall, ViT+MGSF achieves strong recognition across most categories. Despite its robustness, a few categories remain challenging. The most notable confusion occurs for the category of navigation, with a number of samples misclassified as "water intake and outfall." Mild mutual confusions also occur among structurally related categories, particularly between "impermeable structure" and "cross-sea bridge" or "permeable structure." Although coastal aquaculture and open aquaculture share similar grid-like patterns and water-dominated backgrounds, they are generally well distinguished, with only limited errors caused by variations in facility scale, spatial layout, and surrounding coastal context. These confusions highlight the challenging nature of CROSS-14, in which multiple categories exhibit high interclass similarity.

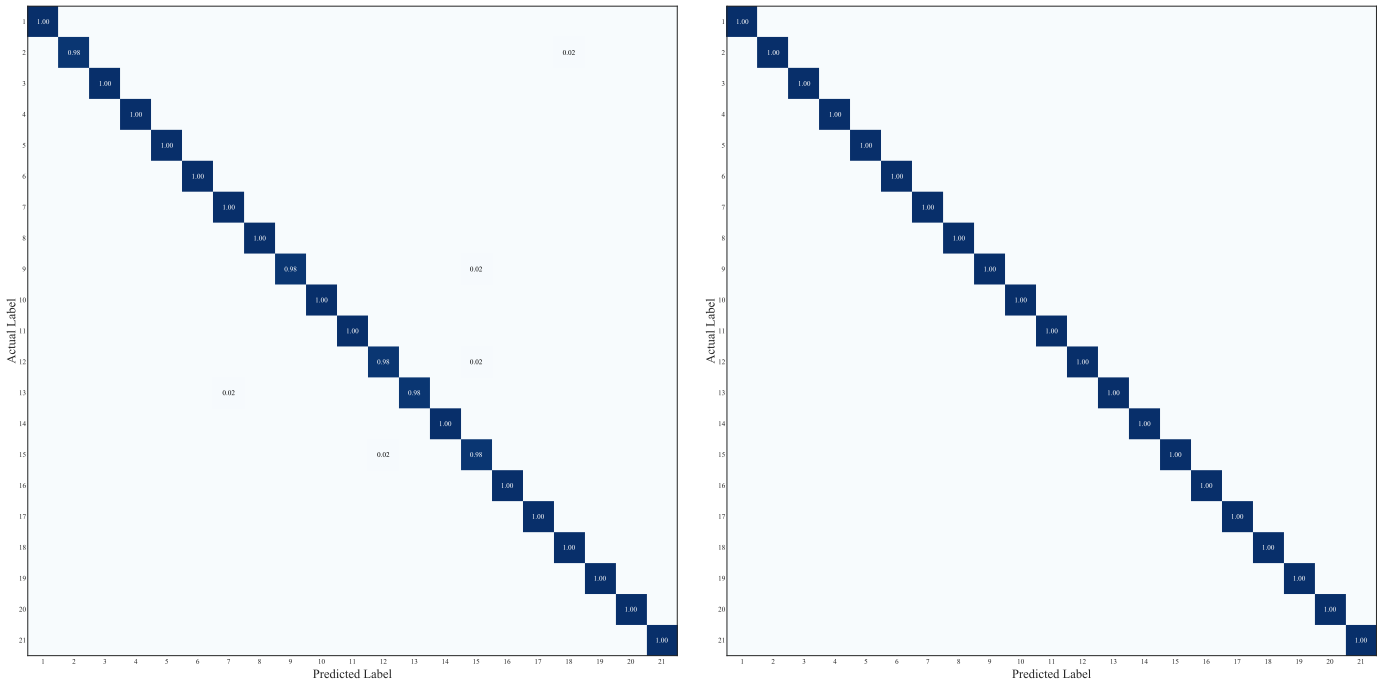
2) Comparison on UCM: The classification results in Table II show that the proposed MGSF-based models achieve the best performance over all compared methods under both training settings. When we use 50% training samples, ViT+MGSF achieves an OA of 99.43%, outperforming VGG16, ResNet50, Swin-T, AdaptFormer,

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(a) (b)

Fig. 4. CMs of ViT+MGSF on the CROSS-14 dataset with (a) 50% and (b) 80% training ratios. The class indices correspond to the category order listed in Section III-A.

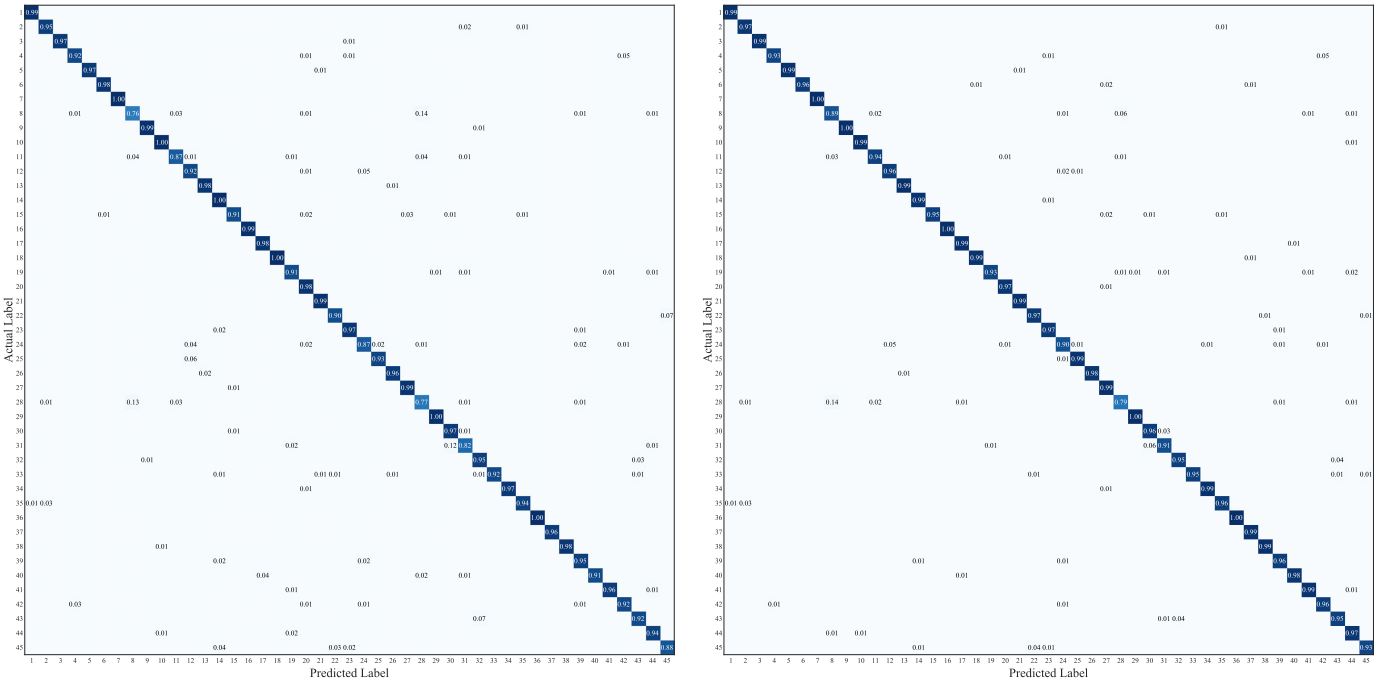


(a) (b)

Fig. 5. CMs of ViT+MGSF on the UCM dataset with (a) 50% and (b) 80% training ratios. The class indices correspond to the category order listed in Section V-A.

PoolFormer, and FPET, which achieve 96.18%, 96.88%, 99.14%, 99.02%, 99.29%, and 99.04%, respectively. Compared with five RSSC methods, ViT+MGSF outperforms CDLNet, EMSCNet, MSCN, CGFNet, and EMTCAL by 0.37%, 0.33%, 0.17%, 0.95%, and 0.76%, respectively. With the training ratio increased to 80%, ViT+MGSF achieves the highest OA of 99.90%, while the best performance among all compared methods is 99.80%. Meanwhile, consistent improvements can also be observed for DeiT+MGSF and T2T-ViT+MGSF, both of which outperform their correspond-

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(a) (b)

Fig. 6. CMs of ViT+MGSF on the NWPU dataset with (a) 10% and (b) 20% training ratios. The class indices correspond to the category order listed in Section V-A.

ing backbones and compared methods. These results further demonstrate the effectiveness of the proposed MGSF framework on small-scale datasets, where overfitting and limited training samples often pose challenges.

Fig. 5 shows the CMs of ViT+MGSF. At a 50% training ratio, ViT+MGSF correctly classifies nearly all categories, with only a small number of samples from airplane, freeway, intersection, and overpass being misclassified. When 80% of the data is used for training, all classes are correctly classified. These results further confirm the effectiveness of MGSF in enhancing classification performance on UCM.

3) Comparison on NWPU: In Table II, the VFMs fine-tuned by MGSF also produce better evaluation results than compared methods. With 10% training samples, ViT+MGSF achieves the highest OA of 94.54% and outperforms Swin-T, VGG-16, ResNet50, AdaptFormer, PoolFormer, and FPET. Vit+MGSF also outperforms all five RSSC methods by a margin ranging from 0.96% to 2.91%. When 20% of the training samples are used, ViT+MGSF further obtains the best OA of 95.94%, better than CNN- and transformer-based methods. Moreover, DeiT+MGSF and T2T-ViT+MGSF exhibit consistent performance gains over their corresponding backbones and most competing approaches. The NWPU dataset contains a large number of scene categories with complex spatial patterns and high intraclass variability. The strong performance achieved by MGSF indicates its effectiveness in handling challenging large-scale RSSC benchmarks, highlighting its robustness and generalization capability.

The CMs in Fig. 6 show that ViT+MGSF performs well in most categories. However, confusion remains between the palace and church classes, which are frequently misclassified as each other due to their strong visual similarity. For the results trained on 20% training data, the OA of different classes is improved, but the confusion between "palace" and "church" persists.

Although MGSF is motivated by the distinctive characteristics of coastal scenes, its sample-, token-, and channel-granularity selection mechanisms are not restricted to coastal categories. The strong results on UCM and NWPU indicate that MGSF can also serve as a general adaptation framework for RSSC, while its advantages are particularly relevant to coastal scenes with redundant backgrounds and sparse discriminative structures.

D. Ablation Study

In this section, ablation experiments are conducted to investigate the contribution of each granularity selection strategy in the proposed MGSF framework across three RSSC datasets. In particular, the UAR and token- and channel-granularity selection mechanisms are individually removed from ViT+MGSF to assess their respective effects. Table III presents the OA of ViT+MGSF under different configurations. From the results in Table III, it can be observed that incorporating any individual module leads to a noticeable improvement in classification performance compared with the vanilla model. In particular, for the models trained on 50% data, introducing only the UAR strategy results in OA improvement of 0.77%, 0.19%, and 0.25% on the CROSS-14, UCM, and NWPU datasets, respectively. Although the improvement is relatively modest, this result confirms the effectiveness of the UAR strategy in optimizing sample weighting and mitigating data

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TABLE III
ABLATION RESULTS OF VIT+MGSF

UAR	Token selection	Channel selection	CROSS-14		UCM		NWPU	
			50%	80%	50%	80%	10%	20%
✗	✗	✗	73.75±0.13	76.01±0.09	90.29±0.25	92.62±0.11	80.02±0.10	83.21±0.20
✓	✗	✗	74.52±0.19	76.14±0.07	92.48±0.46	95.24±0.18	80.27±0.22	83.55±0.16
✗	✓	✗	86.88±0.07	88.44±0.03	98.22±0.11	99.05±0.12	94.08±0.33	95.42±0.21
✗	✗	✓	86.67±0.21	88.02±0.12	98.10±0.07	99.76±0.04	93.62±0.02	95.52±0.21
✓	✓	✗	86.62±0.07	88.89±0.16	99.14±0.14	99.76±0.25	93.82±0.22	95.60±0.11
✓	✗	✓	85.76±0.22	88.13±0.10	99.33±0.03	99.76±0.18	93.15±0.28	95.56±0.13
✗	✓	✓	86.32±0.15	88.76±0.28	99.21±0.11	99.52±0.16	94.20±0.12	95.64±0.07
✓	✓	✓	**87.13±0.14**	**89.02±0.11**	**99.43±0.13**	**99.90±0.10**	**94.54±0.10**	**95.94±0.11**

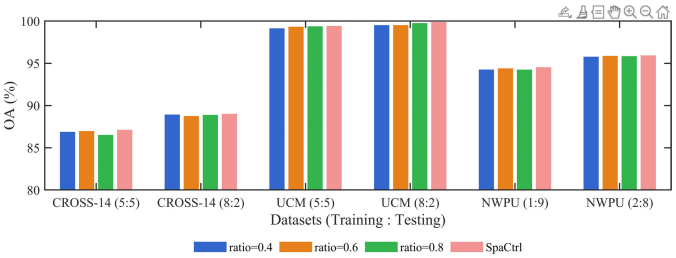


Fig. 7. OA values of ViT+MGSF with different retention ratios of channels and tokens on three datasets.

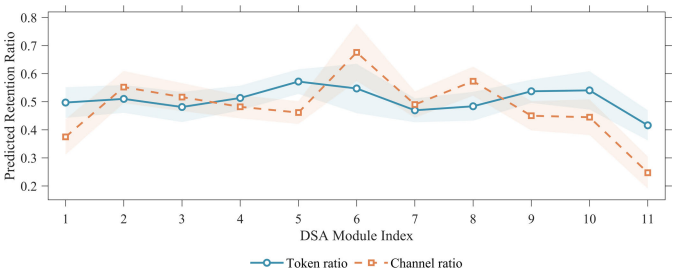


Fig. 8. Predicted token and channel retention ratios of SpaCtrl across different DSA modules on the CROSS-14 test set under the 80% training setting. The curves represent the mean values over all test samples, and the shaded regions indicate the corresponding standard deviations.

imbalance. On the contrary, both the token- and channel-granularity selection strategies bring substantial performance gains. When the channel-granularity selection is employed alone, the OA improves by 12.92%, 7.81%, and 13.6% on the three datasets, respectively. Similarly, using only the token-granularity branch results in OA gains of 13.13%, 7.93%, and 14.06%. These results indicate a slight advantage of token-granularity selection over channel-granularity selection in improving the discriminative representation of extracted features due to its stronger ability to emphasize informative regions. The integration of these two granularity selection mechanisms significantly enhances the capability of VFMs to capture fine-grained contextual relationships in the coastal RSSC task. For MGSF, the best results are achieved when all three modules are combined. The three components provide complementary benefits at different granularities. UAR emphasizes difficult samples during optimization, while token and channel selection highlight discriminative spatial regions and informative feature dimensions within each sample, respectively. Their combination therefore improves both sample-level optimization and feature-level adaptation.

E. Analysis of Channel and Token Granularities

1) Effect of SpaCtrl: To further investigate the impact of the selection of channels and tokens in the DSA module within ViT+MGSF, we replace the retention ratio dynamically predicted by SpaCtrl with a fixed one. In particular, the retention ratio is manually set to 0.4, 0.6, and 0.8, where small values mean higher sparsity. The OA under different settings is reported in Fig. 7. As the CROSS-14 dataset contains large homogeneous backgrounds and small objects, the OA decreases as the retention ratio increases from 0.4 to 0.8. This indicates that a small ratio is more effective at removing redundant background tokens and highlighting discriminative regions. On the contrary, the accuracy on the UCM dataset increases when more tokens or channels are preserved, with a retention ratio of 0.8 yielding the best performance among the fixed-ratio settings. For the NWPU dataset, the model achieves the best accuracy at a moderate sparsity level, suggesting that an appropriate balance between redundancy reduction and information retention is crucial in this dataset. More importantly, the retention ratios dynamically predicted by SpaCtrl consistently achieve the best OA values compared to the manually specified ratios. In other words, the learned retention ratios enable MGSF to automatically arrive at near-optimal sparsity levels of different datasets, which validates the effectiveness of the proposed SpaCtrl.

To further examine the adaptive behavior of SpaCtrl, we record the token and channel retention ratios predicted by all DSA modules. As shown in Fig. 8, both the mean ratios and their standard deviations vary across modules, indicating layer- and sample-dependent sparsity. This adaptive prediction enables SpaCtrl to better match the feature characteristics of different layers and samples than a fixed retention ratio.

2) Effect of ImpEst: To further investigate the effect of ImpEst in the DSA module, we replace the importance scores generated by ImpEst with randomly generated scores. In particular, the random-score variant keeps the same channel and token selection operations while only changing the source of the importance scores. The OA values under different settings are reported in Fig. 9. It can be observed

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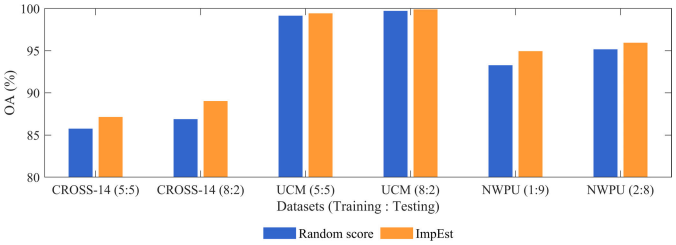


Fig. 9. OA values of ViT+MGSF with different importance score generation strategies of channels and tokens on three datasets.

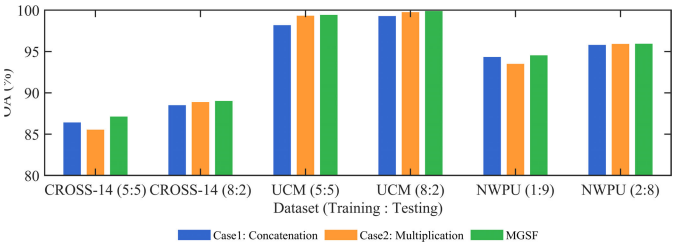


Fig. 10. OA values of ViT+MGSF with different fusion strategies of channels and tokens on three datasets.

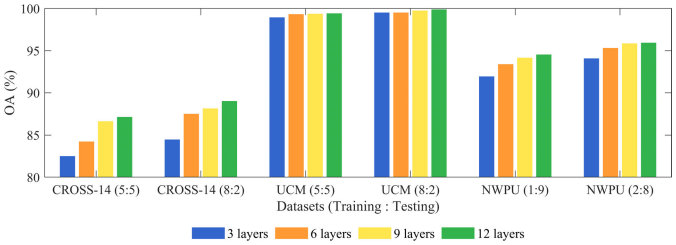


Fig. 11. Impact of the embedding position of DSA within ViT on three datasets.

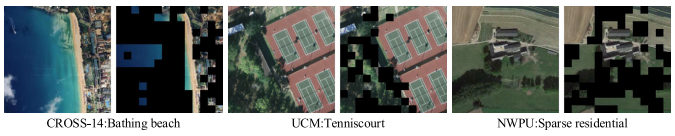


Fig. 12. Visualization of token selection results on the CROSS-14, UCM, and NWPU datasets.

that ImpEst consistently achieves better performance than the random-score variant on all three datasets. This indicates that the learned importance scores are more effective in identifying informative channels and tokens. On the CROSS-14 dataset, the performance gap is more obvious, suggesting that adaptive importance estimation is particularly useful for coastal scenes with large homogeneous backgrounds and sparse foreground objects. On the UCM dataset, although both settings obtain relatively high accuracy, ImpEst still brings stable improvements. On the NWPU dataset, ImpEst also achieves better results under different training ratios, further demonstrating its effectiveness for complex large-scale scene classification. These results show that the improvement of DSA does not simply come from random feature sparsification but from meaningful importance estimation of channels and tokens.

3) Fusion of Selected Channels and Tokens: The channel-refined query Q_c and token-refined query Q_t encode

TABLE IV

HYPERPARAMETER ANALYSIS OF UAR ON CROSS-14 UNDER DIFFERENT TRAINING RATIOS

t_1	t_2	α	β	CROSS-14	
				50%	80%
2.0	1.0	-0.1	0.10	87.03±0.10	87.88±0.12
1.8	1.35	-0.2	-0.05	86.11±0.19	88.13±0.21
2.2	1.65	-0.3	-0.05	86.44±0.07	88.51±0.16
2.0	1.5	-0.3	-1.0	86.98±0.22	88.38±0.15
2.0	1.5	-0.1	0.0	86.06±0.11	88.38±0.08
2.2	1.65	-0.1	0.0	86.62±0.14	88.89±0.16
1.8	1.35	-0.3	-0.10	86.16±0.09	88.76±0.14
2.0	**1.5**	**-0.2**	**-0.05**	**87.13±0.14**	**89.02±0.11**

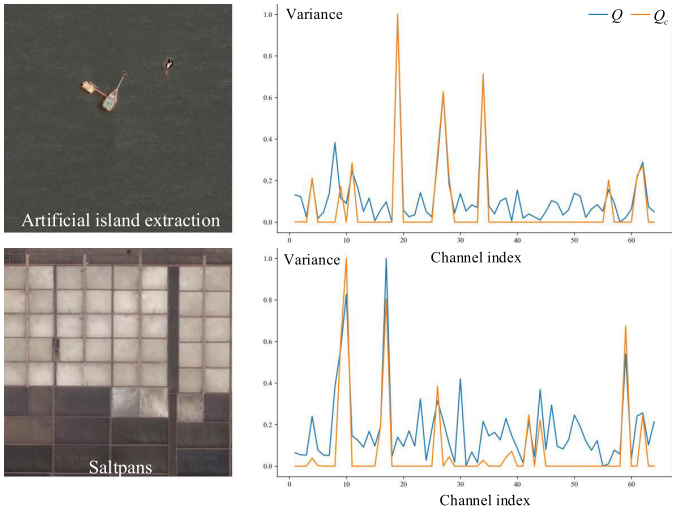


Fig. 13. Visualization of channel selection results on the CROSS-14 dataset.

complementary information corresponding to the selected channels and tokens, respectively. To achieve a more expressive feature representation and improved classification performance, it is essential to effectively fuse the two components. To validate the effectiveness of the fusion strategy in (8), two different cases are compared. In the first case, Q_c and Q_t are directly concatenated and mapped by a 1×1 convolution layer for channel adjustment. In the second case, W_c and W_t are combined with Q via right and left matrix multiplications for the generation of Q_f. Fig. 10 illustrates the performance of different strategies on all datasets. As shown in Fig. 10, the adopted strategy in (8) produces the best values. In the first scheme, the concatenation operation preserves the complementary information of kept channels and tokens. In the second scheme, it may increase the sparsity of the query when both left and right multiplications are applied, leading to a possible loss of important information.

F. Embedding Position of DSA

To further investigate the impact of embedding DSA at different network depths, we insert it into the last three, six, and nine layers of ViT, as well as into all layers, for comparison. As illustrated in Fig. 11, the classification accuracy of

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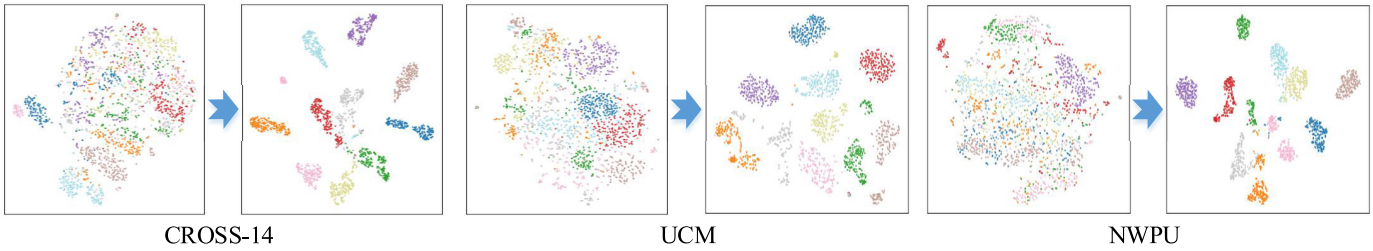


Fig. 14. t-SNE visualization of feature distributions before and after channel selection.

ViT+MGSF gradually improves with the increasing number of embedded layers. When DSAs are inserted in all layers, the model achieves the highest accuracy across all datasets. These results show that progressively refined representations are achieved at both token and channel granularities when more DSAs are equipped. Accordingly, we adopt the full-layer embedding strategy in the final model to maximize the effectiveness of DSA.

G. Hyperparameter Analysis

The UAR function assigns sample weights according to prediction correctness and uncertainty. The temperature coefficients t_1 and t_2 determine the entropy sensitivity in the correctly classified and misclassified branches, respectively, while α and β adjust the overall weight scale. Table IV summarizes the grouped hyperparameter study on CROSS-14 under 50% and 80% training ratios. The best performance is obtained with $(t_1, t_2, \alpha, \beta) = (2.0, 1.5, -0.2, -0.05)$, achieving 87.13% and 89.02% OA, respectively. The performance is stable under moderate perturbations, whereas excessively high or low settings yield slight accuracy drops, suggesting that a balanced entropy sensitivity and weight scale are preferable.

With $t_1 = 2.0$ and $\alpha = -0.2$, correctly classified low-entropy samples are explicitly downweighted to suppress redundant learning, while the weights of correct samples with high uncertainty are close to 1 to facilitate decision-boundary refinement. With $t_2 = 1.5$ and $\beta = -0.05$, misclassified samples are upweighted in a controlled manner, imposing stronger penalties on high-confidence errors while avoiding unstable optimization.

H. Visualization

To qualitatively verify the effectiveness of DSA in terms of the selection of channels and tokens, we first visualize the remaining tokens in Fig. 12. As shown in the figure, we can see that the selected tokens can identify the discriminative spatial regions, and most homogeneous background regions are filtered out. This observation confirms that the proposed token-selection mechanism successfully concentrates attention on informative foreground structures. We further visualize the selected channels in Fig. 13, where the variance of each channel is computed to reflect its relative importance. The figure shows that the channels with high variance in the original query are kept in the refined query, whereas those with low variance are suppressed. This result demonstrates

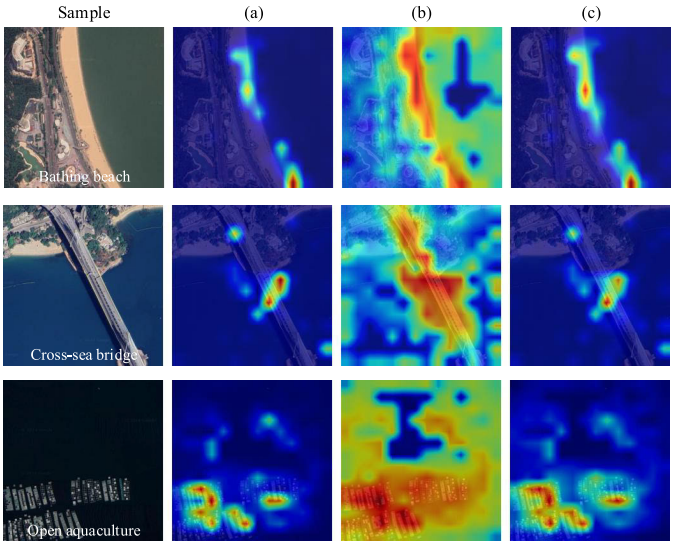


Fig. 15. Visualizations of images on CROSS-14 dataset. (a) Token granularity. (b) Channel granularity. (c) Adaptive fusion.

that channel-granularity selection plays a complementary role in enhancing feature discrimination for the RSSC task.

To further validate the effectiveness of the retained channels, we perform a t-distributed stochastic neighbor embedding (t-SNE) visualization on three datasets. In particular, for each dataset, we randomly select one image from ten scene classes and treat each token of the feature of the last layer as an individual data point. Consequently, each image corresponds to N points in the visualization results. Then, the features with all channels and selected channels are projected into a 2-D space using t-SNE, respectively. Fig. 14 shows the visualization results, which reveal that the suppression of redundant channels results in more clustered features. After imposing ImpEst and SpaCtrl on features, the reserved channels exhibit a much more compact distribution. These observations further confirm that the proposed channel-selection mechanism effectively enhances feature representation quality.

Finally, to provide an overall analysis of the dual-granularity selection mechanism, Fig. 15 illustrates the heat maps of token granularity, channel granularity, and dual-granularity selection. As shown in Fig. 15, token-granularity selection and weighting primarily focus on spatial tokens with high confidence, thereby emphasizing discriminative local structures such as salient regions and object boundaries. In the CROSS-14 examples, these regions correspond to beach-water boundaries, bridge

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bodies, and aquaculture facilities, while homogeneous water backgrounds receive weaker responses. On the contrary, the channel selection mechanism performs filtering along the channel dimension without removing any tokens, resulting in smoother and more globally consistent responses. As shown in the heat map of the dual-granularity selection, local sensitivity and global consistency are balanced as the responses in salient target areas are enhanced while the continuity of the background is maintained. These qualitative observations are consistent with the ablation results in Table III, where token- and channel-granularity selection produce substantial OA improvements across all three datasets.

VI. Conclusion

In this article, we systematically investigate the coastal RSSC task, which is characterized by large homogeneous backgrounds and sparsely distributed foreground facilities. To facilitate fine-grained coastal scene understanding, we construct a dedicated coastal RSSC dataset covering 14 coastal use patterns. Building upon this dataset, we propose MGSF, a framework that adapts VFMs through progressive selection at the sample, token, and channel granularities. By incorporating UAR and DSA, discriminative information is preserved while the redundant representations are suppressed. Extensive experiments on CROSS-14 and two public benchmarks demonstrate that MGSF achieves better performance compared with state-of-the-art methods. Moreover, ablation studies and in-depth analysis also confirm the effectiveness of the proposed multigranularity selection mechanisms and their roles in terms of enhancing representation learning. Despite the promising results, coastal RSSC remains a challenging problem, particularly for categories with strong visual similarity. Another limitation is that the sample difficulty estimation in UAR relies on the predictions obtained from linear probing, and inaccurate predictions may affect the subsequent sample weighting. In addition, the current CROSS-14 dataset is collected exclusively from the Chinese coastal zone. For future work, we plan to explore more robust difficulty estimation strategies, construct a larger global coastal RSSC dataset with broader geographic coverage and richer categories and further improve classification accuracy by boosting the discriminability of VFMs on visually similar coastal use patterns.

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